

Project Topic: An Adaptive “Self-Healing” SDN Framework for Mitigating Control Plane Attacks

Project Plan and Goal

Primary Goal: Address the “Single Point of Failure” vulnerability in SDN architecture. By monitoring Packet_In message rates and controller CPU health, the framework will identify DDoS flooding and switch to more dynamic flows within the data plane to prevent total controller failure.

Baseline Objectives:

1. Developing a controller monitoring scheme for throttling and blocking malicious traffic
2. Implement a “restoration phase” that will automatically remove restrictive flows after threat levels drop to restore full network service for legitimate users.
3. System should prevent the controller from exceeding certain utilization thresholds during a sustained 10,000 packet-per-second flood

Schedule / Work distribution

Week	Task	Member Responsible
1	Literature Review and CloudLab Setup	Vicki & Edward
2/3	Writing XML/Python to define distributed topology	Vicki
2/3	Baseline Performance Benchmarking: Stress-testing CloudLabs to identify a saturation point	Edward
4	Distributed Detection Development: System Monitor/Analyzer	Edward
5	Rigging Detection Logic to Flow-mod pusher to automate DROP and RATE-LIMIT commands	Vicki & Edward
6	Self-healing and Restoration Logic that cleans up mitigation rules	Vicki
7	Data Synthesis and Final Report	Vicki & Edward

Test plan & Environment

Test Plan:

1. Normal traffic baseline behavior
2. Packet_In saturation attack (no protection)
3. Packet_In saturation attack (with protection enabled)
 - a. Granularity Test: Allowing valid clients to maintain similar throughput during attacks
4. Post-attack recovery validation

Environment:

- CloudLab
- Operating System: Ubuntu 22.04
- SDN Stack: Ryu Controller and Open vSwitch
- Traffic Generation: Scapy